

A Conversational Shopping Assistant for Online Virtual Stores

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Figure 1: (a) conversational assistant; (b) combine outfits in mannequin; (c) a store section.

ABSTRACT

The online shopping industry benefits from the usage of virtual assistants that are able to provide a 24/7 way of communication. This is normally done through the use of text based interactions. In order to improve the user experience and provide a more engaging user-to-customer interaction, we present a virtual shopping environment that emulates the physical store, providing the user with an experience similar to buying products in the actual store. This allows users to interact with the store items and have a full conversation with a virtual shopping assistant. This is all done through a virtual store where the user is able to walk, try-on and see the displayed items, while having a conversation with the store's assistant.¹

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**.

KEYWORDS

Virtual Shopping, 3D Conversational Assistant, Multimodal Conversational Assistant

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¹Visual Demo: <https://drive.google.com/file/d/1zfHPwAaSSGD5KR7fDEfAzTVTedr6CNiv/view?usp=sharing>

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1 INTRODUCTION

This paper describes a demo of a 3D virtual shopping environment. The shopping environment is centred in a virtual room that emulates the experience of buying products in physical stores. The user will start in the virtual world at the store entrance, where a conversational assistant will be welcoming and guiding the user through the shopping experience, Figure 1. The application is composed by the following component:

- **Virtual Store.** The virtual environment representing the store considers the usability and user experience as a key factor [1]. The store has several sections with different items - this is crucial to organise products into clear sections, so users can intuitively navigate the store. To achieve this, it is important that we follow the usability heuristic "*Aesthetic and Minimal Design*" [1]. Another heuristic used in the development of this interface is the "*Flexibility and Efficiency of Use*" [1]: the interface should provide an efficient and easy way for the user to complete tasks. Effort should be reduced to achieve the expected results.
- **Conversational Assistant.** The virtual assistant can communicate with the user through a conversation with cross-references to both visual traits and metadata of products.
- **Visual Product Rendering.** As this interface will work mainly with images of clothing items, it is necessary to pre-process those images so they are all normalised. We also use image processing techniques to generate the back shots of the clothing items so that we are able to generate the 3D clothing models.

2 VIRTUAL SHOPPING ENVIRONMENT

This section describes the key features of the shopping environment. Every feature can be categorised into three distinct sets, the virtual assistant, that is responsible for the providing a communication interface with the chatbot, the store itself, and the products.

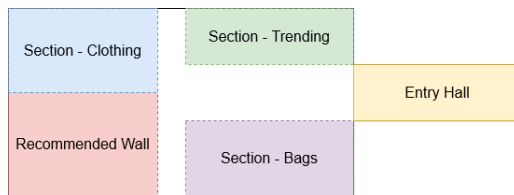


Figure 2: Virtual store layout.

2.1 User-Assistant Interaction

As the goal is to develop a conversational assistant for a clothing store, for most of the clothing items the user will need to choose a size and in some cases the color of the piece. To facilitate this action, when the user textually confirms that they want to add some item to the cart, the assistant will prompt a message composed of images and buttons that inform the user of all the available sizes for that clothing item. This feature is useful for clothing that uses different measurements. For example, some clothes might be marked with number and others with letters (e.g. S, M, L, XL). In this way, the error of the user inserting a value that is not valid is avoided and makes the selection easier and more efficient. Another type of message consist of a list of recommended items. Each item has shortcuts and the user can also click on them to see its full details. This type of message is similar to the one presented in [3].

One last functionality is the ability to input images from the user device, allowing the user to make requests such as looking for clothing that is similar to the uploaded piece. In figure 1 where the user is standing in the entrance hall it can be seen an example of the chatbox with a recommendation, in this case the use of the sizes buttons when the user selects an item to buy.

2.2 Virtual Store

The interface was designed so that there are multiple sections in the store, Figure 2. Each section includes some pre-defined outfits and a set of display screens that show some section items. These can be the current trends (purple zone in Figure 2) or items that are on sale. There are also shelves that are used to display items like jewellery or purses. Each item can be selected to show its details, and provide the user with a set of tasks.

When entering the store, there is an attendant at the *entry hall* (yellow zone in Figure 2) that follows the user around. This is done so to maximize the user's interaction with the chatbot. These features also improve the immersion of the user.

The *recommendation wall* (red zone in Figure 2) is a shared place where the user can see the recommended items and all the ones that were marked as interested. The items are displayed side by side and the users can combine them to create their own outfits (eg. if some pants go well with a t-shirt). This is done by clicking and dragging the desired item to the outfit we intend to create. If the items are t-shirts, trousers or shorts they can be seen in the 3D mannequin.

Upon clicking on a mannequin it will bring up the mannequin details interface, where the user can customise the mannequin's outfit in an easier and quicker way. In the right side, the user can see all the items that are present in the recommended section, divided

in bottom and top clothing parts. Clicking on an item will place it on the mannequin.

2.3 Products Visualisation and Combination

There are two ways that items can be suggested to the user: when talking with the chatbot and giving it a textual description of the desired item, or by providing a piece of clothing and asking for similar clothes, as seen in Figure 1 (a). As mentioned above, those items will appear in the chatbox and will also be added to the store's *recommendation wall*. In Figure 1 we can see an example of the recommended items in the chatbox on the right corner.

For each item that the user can access, it is possible to select it and see its details on the user interface. Here, they can see more pictures of the product, its price, brand and description. The user can also choose a size and add the item to the cart. They can also mark the item as interesting (consequently adding it to the *recommendation wall*) or visit the products website for more information. In Figure 1, we can see the details of an item that we can find in the store outfits.

Product Combination. Following existing work with 3D models [2], these were used to improve the user experience, as they allow the user to see the clothing as she/he would do by wearing them in the store. It is then possible, for example, to check if a shirt looks good with some jeans. The user can easily combine different pieces of clothing with others.

3 USER EXPERIENCE

While communicating with the agent, some items will be recommended to the user. To facilitate the conversation and provide a more engaging interaction, the interface provides different types of messages (text, images and buttons), both for the user and for the agent. Users have the ability to insert an image from their devices with the purpose of asking for similar clothing.

As the user goes through the virtual building, it is possible to visit different sections of the store, that are organised based on the distinct clothing pieces they contain (jackets, jeans, shoes, etc.) or in regard to specific trends and brands.

Each individual section of the store layout has its own item display, and there is also a *trending outfits section* that provides complete outfits to the user (green zone in Figure 2). There are many kinds of displays that vary according to the items that each section contains (e.g. if there are clothing items, jewellery or bags). The user will be able to select each item, ask for more information about it and access the product on a website. The user can also add the selected item to their recommendation section.

The recommended items are then showed in a common section between all other sections, Figure 1(c). The user is also able to combine multiple items into an outfit to see if they go well together. The items are usually seen in 2D view but using the former method they can be visualised in a 3D mode. The items that make part of this section are the ones that the conversational assistant recommends or other section items that the user marked as interested.

To provide a better user experience, the attendant also features several animations, like greeting the user upon entering a section and talking gestures when the chatbot sends a message.

4 CONCLUSION

In this article, we presented a new way of communication with a chatbot for the clothing industry. This approach allows the user to have a more realistic experience, getting closer to the real user-to-costumer interaction, while providing different ways of interacting with the store items.

Experimental results, including preliminary user testing, showed that target users would benefit from an interface such as the one presented here, enabling some degree of embodiment, providing realistic interactions and a better user experience than with text only chatbots.

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